FYS4480/9480 October 15

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HF :

(i) $\varepsilon_p^{HF} = \langle p|\hat{h}_0|p\rangle + \sum_{j\le F} \langle pj|v|pj\rangle_{AS}$

(ii) $\langle \Phi_0^{HF}|\hat{H}|\Phi_0^{HF}\rangle = E_0^{HF}$

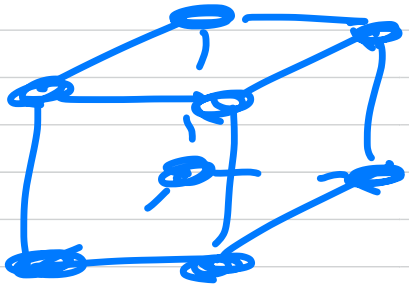
$= \sum_{i\le F} \langle i|h_0|i\rangle + \frac{1}{2}\sum_{ij\le F} \langle ij|v|ij\rangle_{AS}$

(iii) Thouless theorem $|C'\rangle = e^{\hat{T}_1}|\Phi_0^{HF}\rangle$

$$\hat{T} = \sum_{qi} c_n^a a_a^+ a_n$$

Infinite Homogeneous electron gas (HEG)

(1) particles in a box of length L (periodic)



volume $\Omega = L^3$

N - electrons

density

$$n = N/\Omega$$

kinetic energy

$$h_0 = \hat{t} = -\frac{\hbar^2 \nabla^2}{2m}$$

three-dim

$$\vec{k} = \frac{2\pi}{L}\left(n_1 \vec{e}_1 + n_2 \vec{e}_2 + n_3 \vec{e}_3\right)$$

(x) (y) (z)

$$n_i = 0, \pm 1, \pm 2, \pm 3, \ldots$$

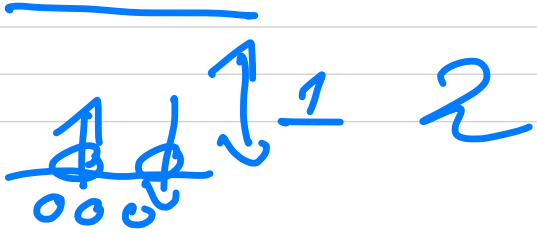
$$\hat{t}\,\varphi_{\vec{k}}(\vec{r}) = \frac{\hbar^2 k^2}{2m}\varphi_{\vec{k}}(\vec{r})$$

$$\varphi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{\Omega}} e^{i\vec{k}\vec{r}}$$

with spin: $\varphi_{\vec{k}}(\vec{r}) \otimes \xi_{\sigma m_\sigma}$

$k^2 \propto n_1^2 + n_2^2 + n_3^2 \Rightarrow$ "magic number"

$n_1^2+n_2^2+n_3^2$	n_1	n_2	n_3	↑↓	
0	0	0	0	2	
1	1	0	0		
	-1	0	0		
	0	1	0	6×2 = 12	14
	0	-1	0		
	0	0	1		
	0	0	-1		
	12 possibilities				38
	-1	-1	0		



Magic numbers:

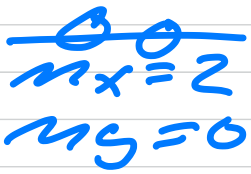
2, 14, 38, 54, - -

Harmonic oscillator in 2-dim

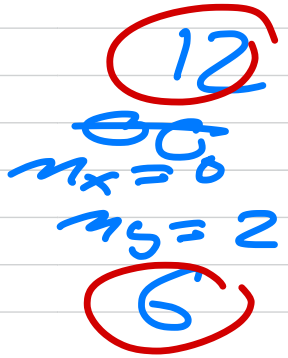
non-interacting energy

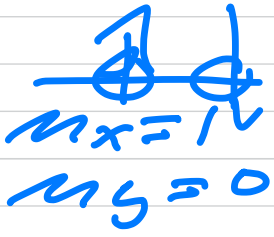
$$E_{n_x n_y} = \hbar\omega\,(n_x + n_y + 1)$$

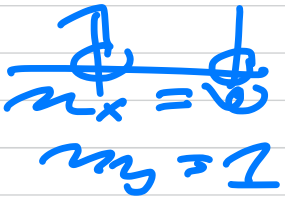
20, 30, 42, - -



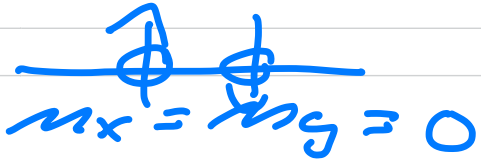








$\hbar\omega$



2

$$\frac{1}{\Omega}\sum_{k\le k_F} \rightarrow \frac{1}{(2\pi)^3}\int_0^{k_F} d^3k$$

$$\vec{k}_i' \propto n_i'$$

$$\Delta k_i' \propto \Delta n_i'$$

$$\sum_{\vec{k}} \rightarrow \int dn_x \int dn_y \int dn_z =$$

$$\left(\frac{L}{2\pi}\right)^3 \int dk_x \int dk_y \int dk_z$$

$$\langle p | u^{HF} | p \rangle = \sum_{j\le F} \langle pj | v | \hat{pj} \rangle - \sum_{j\le F} \langle pj | v | jp \rangle$$

$$\sum_{j \le F} \langle p j | v | p j \rangle$$

$$\hat{V} = \frac{1}{2} \sum_{pq rs} \langle pq | v | rs \rangle a_p^+ a_q^+ a_s a_r$$

$p = (\vec{k}_p, \sigma_p)$ etc

$[1\,0]\,[1\,0]$ $[0\,1]\,[0\,1]$

$$\langle \underbrace{\vec{k}_p \sigma_p}_{(1)} \; \underbrace{\vec{k}_q \sigma_q}_{(2)} | v | \underbrace{\vec{k}_r \sigma_r}_{(1)} \; \underbrace{\vec{k}_s \sigma_s}_{(2)} \rangle$$

v is independent of spin

$$v(\vec{r}_1, \vec{r}_2) = \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$$

$$= \delta_{\sigma_p \sigma_2} \delta_{\sigma_q \sigma_s} \times$$

$$\int d\vec{r}_p \int d\vec{r}_q \; e^{-i\vec{k}_p \vec{r}_p} \; e^{-i\vec{k}_q \vec{r}_p}$$

$$\times \; e^{i\vec{k}_s \vec{r}_q} \; e^{i\vec{k}_n \vec{r}_q} \; v(r) \cdot \frac{1}{\Omega^2}$$

$$\frac{e^{-i\vec{k}_p \vec{r}_p}}{\sqrt{\Omega}}$$

introduce relative coordinate $\vec{r} = \vec{r}_p - \vec{r}_q$, look at integral at fixed r_q

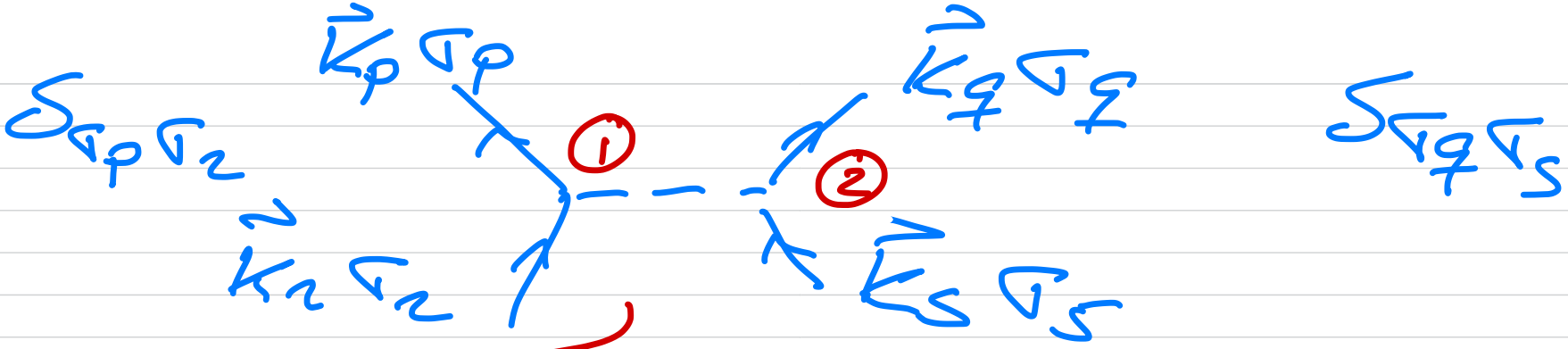
$$= \frac{\delta_{\tau_p \tau_r} \delta_{\tau_q \tau_s}}{\Omega^2} \int v(r) e^{i(\vec{k}_r - \vec{k}_p)\vec{r}} d\vec{r}$$

$$\times \int e^{i(\vec{k}_s - \vec{k}_q + \vec{k}_r - \vec{k}_p)\vec{r}_q} d\vec{r}_q$$

The second integral is a definition of the Kronecker δ which leads to

$$\vec{k}_s + \vec{k}_r = \vec{k}_q + \vec{k}_p$$

conservation of momentum



$$\langle \vec{k}_p \sigma_p \; \vec{k}_q \sigma_q | v | \vec{k}_2 \sigma_2 \; \vec{k}_s \sigma_s \rangle$$

$$\vec{k}_2 + \vec{k}_s = \vec{k}_p + \vec{k}_q$$

collecting

$$\langle \vec{k}_p \sigma_p \; \vec{k}_q \sigma_q | v | \vec{k}_2 \sigma_2 \; \vec{k}_s \sigma_s \rangle$$

$$= \delta_{\sigma_p \sigma_2} \delta_{\sigma_s \sigma_q} \delta_{\vec{k}_2 + \vec{k}_s, \vec{k}_p + \vec{k}_q} \times \int v(\vec{r}) e^{i(\vec{k}_2 - \vec{k}_p)\vec{r}} d\vec{r}$$

$$\varepsilon_p^{HF} = \langle p|\hat{t}|p\rangle + \langle p|u^{HF}|p\rangle$$

$$\langle p|u^{HF}|p\rangle =$$

$$\sum_{j\le F} \langle pj|v|pj\rangle \quad \text{Direct}$$

$$- \sum_{j\le F} \langle pj|v|jp\rangle \quad \text{Exchange}$$

$$= \sum_{k_j\sigma_j} \langle \vec{k}_p\sigma_p(1)\ \vec{k}_j\sigma_j(2)|v|\vec{k}_p\sigma_p(1)\ \vec{k}_j\sigma_j(2)\rangle$$

$$- \sum_{k_j\sigma_j} \langle \vec{k}_p\sigma_p(1)\ \vec{k}_j\sigma_j(2)|v|\vec{k}_j\sigma_j(1)\ \vec{k}_p\sigma_j(2)\rangle$$

$\sigma_j = \pm 1/2$

$$\left(\begin{array}{ll} m_{\sigma_j} = +1/2 \ (\uparrow) & \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ m_{\sigma_j} = -1/2 \ (\downarrow) & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ m_{\sigma_p} = +1/2 & \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{array} \right)$$

$$= 2 \sum_{\vec{k}_j} \langle \vec{k}_p \vec{k}_j | v | \vec{k}_p \vec{k}_j \rangle$$

$$- \sum_{\vec{k}_j} \langle \vec{k}_p \vec{k}_j | v | \vec{k}_j \vec{k}_p \rangle$$

Direct term

$$\langle \vec{k}_p \vec{k}_j | v | \vec{k}_p \vec{k}_j \rangle$$

$$= \frac{1}{\Omega} \int d\vec{r}' \frac{e^2}{|\vec{r} - \vec{r}'|}$$

$$2 \sum_{k_j} \langle \vec{k}_p \vec{k}_j | v | \vec{k}_p \vec{k}_j \rangle$$

$$= \frac{2}{\Omega} \cdot ?$$

$$\frac{1}{\Omega} \sum_{k} \to \frac{1}{(2\pi)^3} \int_0^{k_F} d^3k$$

$$n(\vec{r}) = \sum_{k_i} |\varphi_{\vec{k}_i}(\vec{r})|^2$$

(Density operator)

$$\varphi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{\Omega}} e^{i\vec{k}\vec{r}}$$

$$n(\vec{r}) = \frac{1}{(2\pi)^3} \int_0^{k_F} d^3k$$

$$= \frac{4\pi}{(2\pi)^3} \int_0^{k_F} k^2 dk \qquad \text{include spin}$$

$$2 \cdot \frac{4\pi}{(2\pi)^3} k_F^3/3 = k_F^3/3\pi^2 \left(= \frac{N}{\Omega}\right)$$

Direct term

$$\frac{Ne^2}{\Omega} \int \frac{d\vec{r}'}{|\vec{r} - \vec{r}'|}$$

which is cancelled by an equal term from the background of ions

Exchange term:

$$\vec{k}_j = \vec{k}' \quad \wedge \quad \vec{k}_p = \vec{k}$$

$$-\frac{e^2}{\Omega (2\pi)^3} \int d\vec{r} \int d\vec{r}' \int d\vec{k}'$$

$$\times e^{i(\vec{k}'-\vec{k})(\vec{r}-\vec{r}')} \frac{1}{|\vec{r}-\vec{r}'|}$$

introduce a convergence

$e^{-\mu|\vec{r}-\vec{r}'|}$ and then

calculate integral I and

$\lim_{\mu \to 0^+} I$

$$\vec{x} = \vec{r} - \vec{r}' \quad \wedge \quad x = |\vec{x}|$$

$$\vec{y} = \vec{r}'$$

$$\lim_{\mu \to 0^+} \frac{e^2}{\Omega (2\pi)^3} \int d\vec{k}' \int d\vec{y} \int d\vec{x}$$

$$\times e^{i(\vec{k}' - \vec{k})\vec{x}} \frac{e^{-\mu x}}{x}$$

intermediate step

$$\int d\vec{x}\, e^{i(\vec{k}' - \vec{k}')\vec{x}} \frac{e^{-\mu x}}{x} =$$

$$= \int_0^{\infty} dx\, x^2 \int_0^{\pi} d\theta \sin\theta \int_0^{2\pi} d\varphi$$

$$\times e^{i|\vec{k}'-\vec{k}||\vec{x}|\cos\theta} \frac{e^{-\mu x}}{x}$$

$$= \frac{4\pi}{\mu^2 + |\vec{k}'-\vec{k}|^2}$$

$$\lim_{\mu \to 0^+} = \frac{e^2}{2(2\pi)^3} \int_0^{k_F} d\vec{k}' \int d\vec{\Omega} \frac{4\pi}{\mu^2 + |\vec{k}-\vec{k}'|^2}$$

$$\int d\vec{\Omega} = \Omega$$

$$= \frac{e^2}{2\pi^2} \int_{C}^{k_F} \frac{d\vec{k}'}{|\vec{k}' - \vec{k}|^2}$$

$$= \frac{e^2}{\pi k} \left\{ \frac{k_F^2 - k^2}{2} \ln \frac{|k_F + k|}{|k_F - k|} \right\}$$

$$\varepsilon_k^{HF} = \frac{\hbar^2 k^2}{2m} - \frac{e^2 k_F}{\pi} \left\{ 2 + \frac{k_F^2 - k^2}{k k_F} \ln \frac{|k_F + k|}{|k_F - k|} \right\}$$